

# Brian Hepler, PhD

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## PROFESSIONAL SUMMARY

Data Scientist with a Ph.D. in Mathematics and expertise in network analysis, machine learning, and quantum computing. Recently analyzed 121K+ arXiv papers using advanced network science techniques to uncover collaboration patterns in research communities. Proven ability to translate mathematical theory into production-ready solutions. Seeking data-driven roles leveraging deep technical expertise in Python.

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## TECHNICAL SKILLS

- **Languages:** Python, SQL, LaTeX
  - **ML Libraries:** Scikit-learn, PyTorch, XGBoost, NumPy, SciPy, Pandas, Matplotlib, NetworkX
  - **Tools:** Jupyter, Git, ipywidgets, PostgreSQL
  - **Specializations:** Machine Learning, Statistical Modeling, Quantum Computing, Algebraic Topology, Category Theory, Graph Theory, Representation Theory
  - **Certifications:** [Deep Learning Specialization](#) (deeplearning.ai), [Machine Learning Specialization](#) (deeplearning.ai), [SQL for Data Science](#) (Coursera), [Applied Social Network Analysis](#) (Coursera)
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## PROFESSIONAL EXPERIENCE

### Quantum Formalism — *Remote*

#### Mathematics Consultant

Nov. 2024 – Present

- Engineered hands-on computational exercises in Python notebooks demonstrating quantum and classical ML concepts using Scikit-learn, PyTorch, PennyLane, and SciPy.
- Designed and delivered advanced graduate-level curriculum bridging Lie Theory with practical Quantum Computing and Machine Learning applications (Course: [\[Lie Groups with Applications\]](#)).
- Developed visualization tools for dynamic simulation of quantum systems using Matplotlib and ipywidgets; applied concepts like Hamiltonian evolution,  $SU(4)$  decomposition, and error quantification.

### Institut de Mathématiques de Jussieu-Paris Rive Gauche — *Paris, France*

#### Postdoctoral Researcher

2023 – Sept. 2024

- Developed advanced mathematical models for complex systems exhibiting irregular behavior as part of Fondation Sciences Mathématiques de Paris-funded project under François Loeser.

### University of Wisconsin–Madison — *Madison, WI*

#### Visiting Assistant Professor

2019 – 2023

- Developed and published a novel theoretical framework (Publ. RIMS Kyoto Univ.) extending advanced analytical techniques to model complex systems exhibiting irregular behavior.
  - Secured \$5,000 AMS-Simons research grant for reformulating 40-year-old mathematical problem integrating analytical and topological perspectives; designed and taught undergraduate/graduate mathematics curricula.
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## SELECTED PROJECTS

Math Research Compass: arXiv Trends Dashboard (GitHub [\[link\]](#), [mathresearchcompass.com](https://mathresearchcompass.com))

- Built full-stack web application analyzing 121,391 mathematics research papers with advanced topic modeling pipeline (BERTopic, UMAP, HDBSCAN) and Claude AI-enhanced topic labeling across 1,938 discovered research themes.
- Engineered optimized SQLite database architecture with indexed queries and LRU caching, achieving 5x performance improvement in interactive response times for real-time filtering and exploration.
- Developed and deployed professional Shiny dashboard on Heroku with custom domain, featuring dynamic filtering, topic exploration, and author analytics—demonstrating full-stack capabilities from data pipeline to production deployment.

#### **Scientific Collaboration Networks: Multi-Metric Analysis of arXiv Communities** (GitHub [\[link\]](#))

- Analyzed 121,391 papers across 1,938 topics using network science techniques (degree centralization, modularity, small-world coefficients, coreness ratios) to reveal how research communities self-organize through collaboration patterns.
- Discovered fundamental dichotomy: popular fields exhibit modular structures (high modularity, low centralization) while niche areas show hierarchical organization through high coreness ratios and degree assortativity, quantifying how scientific communities scale differently.
- Integrated structural holes analysis (Burt's constraint/effective size) with BERTopic embeddings to identify knowledge brokers and collaboration opportunities across mathematical research domains.

#### **Hamiltonian Simulation: TFIM via Trotterization** (GitHub [\[link\]](#))

- Simulated time evolution of Ising models using matrix exponentiation and Trotter decomposition.
- Visualized quantum observable dynamics and linked numerical error to Lie bracket theory.
- Analyzed the superior convergence behavior of second-order Trotter methods, demonstrating the  $O(t^3/N^2)$  vs  $O(t^2/N)$  error scaling predicted by the Baker-Campbell-Hausdorff formula.

#### **KAK Decomposition and Quantum ML** (GitHub [\[link\]](#))

- Implemented Khaneja-Glaser algorithm for two-qubit gate decomposition and benchmarked against variational learning models.
- Demonstrated practical use of Cartan decomposition and automatic differentiation in PennyLane.
- Achieved near-perfect unitary approximation (~98% fidelity) using a parameterized quantum circuit that mirrors the mathematical structure of the Cartan decomposition for  $SU(4)$ .

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#### **SELECTED PUBLICATIONS** (Google Scholar [\[link\]](#))

- Hepler, Brian. "Modular versus Hierarchical: A Structural Signature of Topic Popularity in Mathematical Research", arxiv e-prints. DOI: [10.48550/arXiv.2506.22946](https://doi.org/10.48550/arXiv.2506.22946)
  - Hepler, Brian and Hohl, Andreas. "Moderate Growth and Rapid Decay Nearby Cycles via Enhanced Ind-Sheaves", Publ. RIMS Kyoto Univ., 61 (2025), no. 1, pp. 1-51. DOI: [10.4171/prims/61-1-1](https://doi.org/10.4171/prims/61-1-1)
  - Hepler, Brian. "Deformation Formulas for Parameterized Hypersurfaces", Annales de l'Institut Fourier, Volume 74 (2024) no. 3, pp. 1153–1188. DOI: [10.5802/aif.3613](https://doi.org/10.5802/aif.3613)
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#### **EDUCATION**

**Ph.D., Mathematics** — Northeastern University, 2019  
*Specialization: Algebraic Topology, Computational Geometry*

**M.S., Mathematics** — Northeastern University, 2014

**B.A., Mathematics** — Boston University, 2012  
*Cum Laude with Distinction in Mathematics*